# Network design and implementation report

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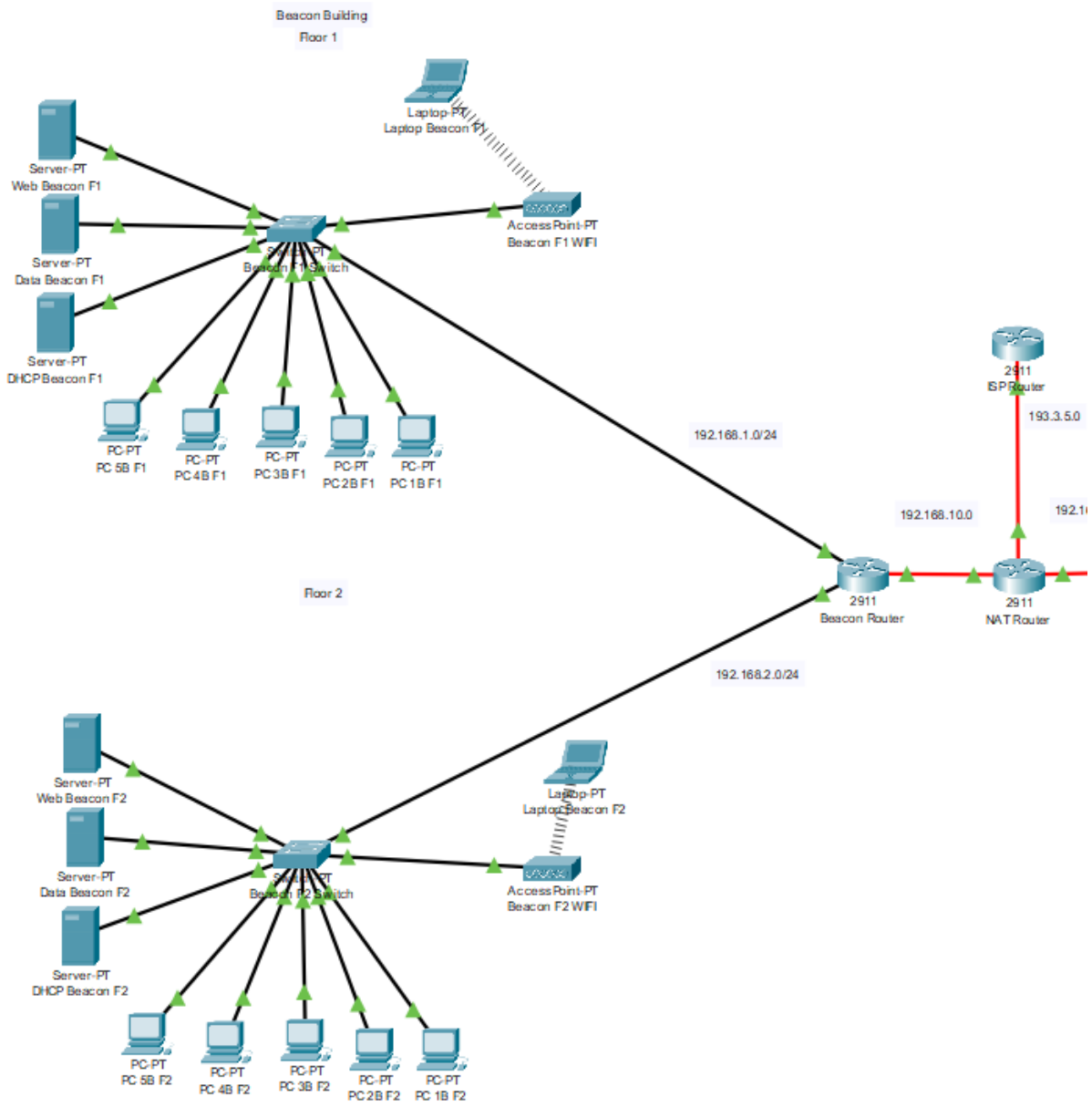
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# Overview and planning

The objective was to design and create a functional network for an electric car manufacturing company that recently acquired two buildings, being the Beacon and Horizon building which each has two floors. Each floor contains a department, requiring isolated network segments and each segment must include wired and wireless connectivity, dedicated servers and internet access through NAT gateways.

Each floor was built with 5 desktop PCs, 1 data, DHCP and web server and a wireless access point which was connected to a wireless laptop. Fibre optic cables were used to interconnect the buildings and the network was connected to the public internet using NAT and an ISP router.




# Device selection and IP scheme

Ive selected Cisco 2911 routers and switch PTS as core networking equipment due to their capabilities to handle multiple VLANs, routing and NAT. Class C private IP address ranges were assigned to each network segments as followed:

| Floor | Subnet | Gateway IP |
|---|---|---|
| Beacon Floor 1 | 192.168.1.0/24 | 192.168.1.1 |
| Beacon Floor 2 | 192.168.2.0/24 | 192.168.2.1 |
| Horizon Floor 1 | 192.168.3.0/24 | 192.168.3.1 |

| | | |
|---|---|---|
| Horizon Floor 2 | 192.168.4.0/24 | 192.168.4.1 |

Two intermediate subnets were used to connect the building routers to the NAT router being:

Beacon to NAT: 192.168.10.0/24

Horizon to NAT: 192.168.20.0/24

NAT was configured to translate internal traffic to public IP being 199.3.5.2, and the ISP router was configured to 199.3.5.1

# Configuration of wired devices and routing

- Static Ips were manually configured for all wired PCs, servers and router interfaces
- Laptops were configured to connect to the wifi on their respective building and floors
- Default gateways on PCs matched their router interface Ips
- On each building router, a default static route was set pointing to the NAT routers interface using the codes:
    - ip route 0.0.0.0 0.0.0.0 192.168.10.1 (Beacon Building)
    - ip route 0.0.0.0 0.0.0.0 192.168.20.1 (Horizon Building)
- On the NAT router, satic routes were added for all internal networks pointing to the building routers

# NAT and ISP configuration

NAT was setup on the NAT router using the following steps:

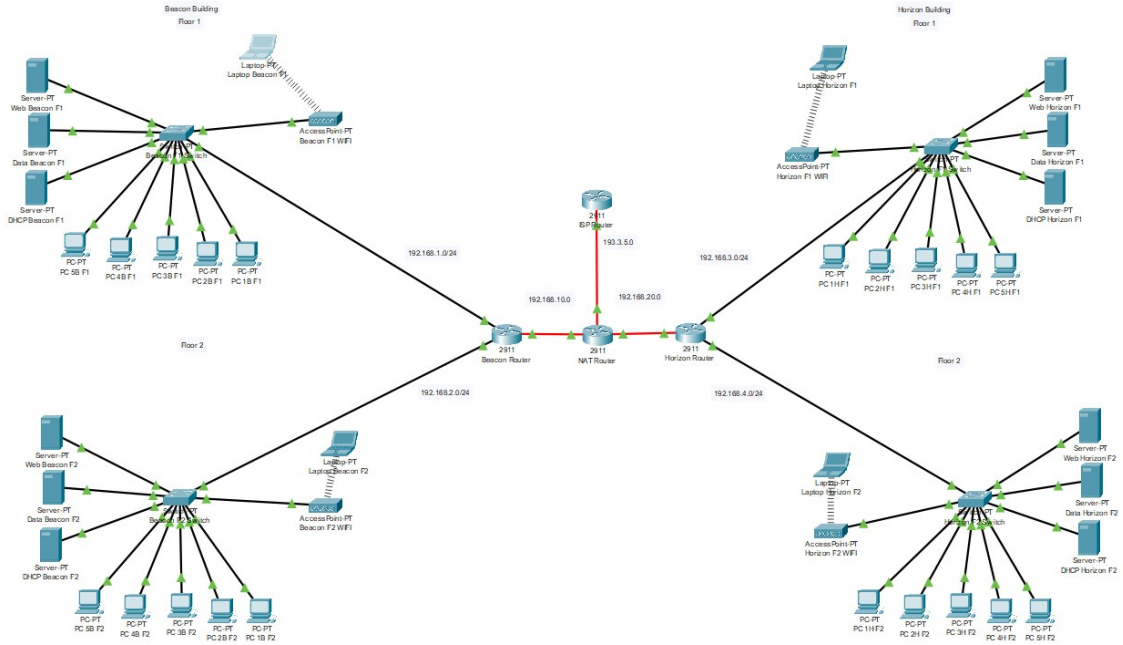
1. Inside interfaces (to beacon and horizon routers) were marked with ip nat inside.
2. The interface toward the ISP was marked with ip nat outside.

The ISP router was configured with a public IP (199.3.5.1) and connected to the NAT routers external interface.

# Wireless Configuration

Each floor had one access point with a unique SSID for example (Beacon F1, Horizon F2) and assigned to a different wireless channel to avoid interference. Laptops were fitted with WPC300N wireless modules, powered on and connected to the matching SSID of the floor that it was on.

# Network connection explanation




## Floor to level connections

Each floor is equipped with:

- Central switch that connects
  - 5 wired PCs
  - 1 data server
  - 1 web server
  - 1 DHCP server
  - 1 wireless access point
- Wireless laptop connecting to access point using WIFI (SSID specific to floor)

## Floor to building router

Each floors switch connects upward to its building router

- Beacon Floor 1 and 2 connects to the beacon router (Cisco 2911)
- Horizon Floor 1 and 2 connects to the horizon router (Cisco 2911)

These routers handle:

- Internal routing between floors
- Default gateway routing for floor devices
- Passing non-local traffic towards the NAT router

## Building to NAT router

The beacon Router and Horizontal Router both connect to the NAT Router using:

- Beacon to NAT: 192.168.10.0/24
- Horizon to NAT: 192.168.20.0/24

The NAT router serves as the central core router that:

- Knows how to route between all internal subnets
- Performs NAT for internet bound traffic

## NAT router to ISP router

The NAT router connects outward to the ISP router over the public subnet:

- NAT external interface: 199.3.5.2
- ISP router: 199.3.5.1

This simulates the companies internet connection.

# Security flaws and countermeasures

## Security Flaw 1: lack of access control on routers and switches

In the current design, routers and switches are left with default configurations, meaning anyone with physical or remote access could potentially log into these devices and change setting. Without console passwords and management access control, the network is vulnerable to configuration tampering, unauthorised routing changes, or even full network shutdowns.

### Countermeasure:

Implement strong password on all network devices. Configure local user accounts or integrate with AAA (Authentication, Authorization and Accounting) services. Additionally disable unused ports on switches to prevent unauthorized device connections.

## Security Flaw 2: Unsecured wireless networks

Each floors wireless access point broadcasts a unique SSID, but if no encryption Is used, attackers with signal range could eavesdrop on traffic, intercept credentials or connect unauthorized devices to the network. This exposes sensitive data and resources to external threats.

### Countermeasure:

Enable strong wireless security protocols (WPA2 OR WPA3) on all access points, with complex passphrases that are regularly changed. Implement MAC address filtering for connected laptops.

## Security Flaw 3: No internal network segmentation for sensitive resources

Currently all PCs, servers and wireless devices on each floor share the same broadcast domain and subnet. If a device on the floor is compromised (from malware

infection or an insider attack) it could directly access sensitive servers like the data or web servers. Theres no isolation between user devices and critical infrastructure.

### Countermeasure:

Implement VLANs to separate user traffic (PCs, Laptops) from server traffic. For example, place all servers into a dedicated server VLAN, configure inter-VLAN routing on the building routers, and apply access control lists to restrict which devices can access the server VLAN. This limits movement across a network if a user device is compromised.

## Overall Recommendations

In addition to the above, apply regular software updates on all network devices, use firewalls at the NAT or ISP edge to filter inbound and outbound traffic, and enable logging and monitoring to detect suspicious activities. Regular security audits should be conducted to ensure all configurations remain compliant with best and newest practices.